Cantera vs NN | Test IC 07 | NRMSE [%]

T0 = 930.6 K P0 = 3.722 atm phi0 = 0.754 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	30.9	26.8	45.6	1.64e+04	24.3	42.4	38.2	24.4	18.7	N/A	Complete
2 P-only DeepONet10	**10k**	38.8	40.7	45.6	31	42.2	41.9	38.1	45.7	40.3	N/A	Complete
2 P-only DeepONet10	**long**	1.54e+07	9.83e+06	1.17e+08	1.19e+10	4.83e+06	4.38e+10	2.74e+07	3.01e+03	3.15e+04	N/A	Complete
3 ANN10 + P-only DeepONet10	**10k**	38.8	40.7	45.6	28.4	42.2	41.9	38.1	45.7	40.3	N/A	Complete
3 ANN10 + P-only DeepONet10	**long**	1.95e+07	1.25e+07	6.74e+07	1.50e+10	6.12e+06	5.36e+10	3.47e+07	5.4e+03	5.82e+04	N/A	Complete
4 T/P DeepONet10	**10k**	39	40.6	1.35e+03	45.3	42.2	66.4	38.1	45.7	40.3	N/A	Complete
4 T/P DeepONet10	**long**	4.20e+06	1.52e+07	2.85e+10	5.83e+09	9.49e+06	1.40e+12	2.07e+07	1.8e+03	3.57e+04	N/A	Complete
5 ANN10 + T/P DeepONet10	**10k**	39.1	38.8	4.81e+03	44.3	39.3	599	37.8	39.6	39.7	N/A	Complete
5 ANN10 + T/P DeepONet10	**long**	5.60e+06	1.81e+06	3.13e+10	1.70e+07	5.16e+04	6.64e+08	1.75e+05	6.71e+03	8.43e+04	N/A	Complete
6 T/P/ϕ DeepONet10	**10k**	2.02e+07	1.45e+07	2.47e+09	1.38e+10	7.52e+06	2.83e+11	3.73e+07	4.91e+03	2.42e+04	N/A	Complete
6 T/P/ϕ DeepONet10	**long**	103	26.2	4.44e+05	27.4	39.7	1.37e+04	30.5	43.5	40.3	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	1.89e+07	1.38e+07	2.39e+09	9.62e+09	7.25e+06	3.16e+11	3.50e+07	5.47e+03	3.63e+04	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	4.12e+05	2.61e+05	4.67e+03	2.97e+04	1.51e+05	1.23e+04	5.87e+05	8.1e+03	1.27e+05	N/A	Complete

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.